



**A Statistical Approach to Evaluating Language Models:
A Comparative Study of GPT-2 and DistilGPT2**

Samyuktha Police

Data Science: Language Resources and Processes

Prof. Ali Ahmed

February 21, 2026

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Abstract

The rapid growth of large language models has greatly improved many tasks in natural language processing, such as text generation, question answering, and conversational systems. As more models become available, it's essential to assess and compare their performance in a consistent and thorough manner. Traditional evaluation methods often rely on basic metrics like accuracy, which only give a rough idea of how a model performs. However, accuracy by itself doesn't always reflect the variability or reliability of the results. Small changes in accuracy between models can happen due to randomness in the dataset or differences in the evaluation process. This can lead to misleading conclusions. For this reason, researchers are increasingly using statistical methods to evaluate machine learning models. A statistical approach to model evaluation looks at not just how well a model performs, but also the consistency and reliability of its results through techniques like variance analysis, confidence intervals, and hypothesis testing.

This paper uses a statistical approach to compare two language models: GPT-2 and DistilGPT2. GPT-2 is a generative language model based on the Transformer architecture, while DistilGPT2 is a smaller version of GPT-2 created through knowledge distillation to cut down on model size and computational costs. In the experiment, both models were tested with a dataset of factual questions. Each model generated responses that were compared to expected answers. The evaluation results were analysed using various statistical metrics, including accuracy, variance, standard deviation, bootstrap confidence intervals, and hypothesis testing. The experimental results indicate that GPT-2 achieved higher accuracy than DistilGPT2, and the statistical analysis confirms that this difference is significant. The study highlights the need for statistical evaluation techniques when comparing machine learning models, as these methods provide a more dependable understanding of model performance beyond simple accuracy measurements.

1. Introduction

In recent years, large language models have emerged as one of the most significant advancements in the field of natural language processing (NLP). These models are designed to understand and generate human language by learning patterns from large collections of textual data. By using deep learning techniques and neural network architectures, language models are capable of performing a wide variety of tasks, including text generation, question answering, language translation, document summarisation, and conversational dialogue systems. Among the many models developed in this area, GPT-2 has become one of the most widely recognised generative language models due to its ability to produce coherent and contextually relevant text. Trained on large datasets of internet text, GPT-2 learns statistical relationships between words and sentences, allowing it to generate responses that resemble human writing. As a result, language models like GPT-2 are increasingly used in applications such as chatbots, virtual assistants, automated content generation, and intelligent search systems.

Despite the impressive capabilities of modern language models, evaluating their performance remains a challenging task. As the number of models continues to grow, researchers need reliable methods to compare different models and determine which one performs better for a particular task. In many machine learning studies, models are evaluated using simple performance metrics such as accuracy, precision, or error rate. While these metrics provide useful information about how well a model performs, they do not always provide a complete understanding of model behaviour. For example, small differences in accuracy between two models may not necessarily indicate that one model is truly better than the other. Such differences may occur due to randomness in the dataset, variations in sampling, or other experimental factors. Consequently, relying only on a single metric such as accuracy may lead to misleading conclusions when comparing machine learning models.

To address these challenges, researchers increasingly apply statistical methods when evaluating and comparing machine learning systems. A statistical approach to model evaluation focuses not only on the average performance of a model but also on the variability and reliability of the results. Statistical techniques allow researchers to analyse how consistent a model's performance is across different samples of data and whether observed differences between models are statistically meaningful. Methods such as variance analysis and standard deviation help measure the stability of the results, while confidence intervals provide an estimated range in which the true performance of the model is likely to lie. In addition, hypothesis testing is commonly used to determine whether the difference between two models is statistically significant or simply caused by random variation in the data. By applying these statistical techniques, researchers can obtain a more rigorous and reliable evaluation of model performance.

Another important aspect of modern language model research is the trade-off between model performance and computational efficiency. Large language models often contain hundreds of millions or even billions of parameters, which allows them to achieve high levels of performance on complex language tasks. However, these large models also require

substantial computational resources, including powerful hardware, large memory capacity, and significant processing time. This can make them difficult to deploy in practical environments where computational resources may be limited. To address this problem, researchers have developed techniques that aim to reduce model size while preserving as much performance as possible. One such technique is model distillation, which involves training a smaller model to mimic the behaviour of a larger model. DistilGPT2 is an example of a distilled model that was created from GPT-2 using this approach. By compressing the original model, DistilGPT2 becomes faster and more efficient to run while attempting to maintain similar language generation capabilities.

The main objective of this term paper is to investigate a statistical approach to evaluating language models by comparing the performance of GPT-2 and DistilGPT2. The study focuses on the question of whether a smaller distilled model can achieve performance comparable to the original larger model. Specifically, the following research question is addressed: Can a smaller distilled model achieve performance comparable to the original larger model? To answer this question, both models are evaluated using a dataset of factual questions that test the ability of the models to generate correct information. The responses generated by the models are analysed using several statistical evaluation metrics, including accuracy, variance, standard deviation, bootstrap confidence intervals, and hypothesis testing. These metrics provide a more comprehensive view of model performance and allow a more reliable comparison between the two models.

The remainder of this paper is organised as follows. Section 2 presents the theoretical background of statistical approaches used for evaluating language models and discusses the statistical metrics used in the analysis. Section 3 introduces the language models used in the study and explains their main characteristics. Section 4 describes the experimental methodology and implementation of the evaluation process. Section 5 presents the experimental results and the statistical analysis of the findings. Section 6 discusses the implications of the results and their relevance to language model evaluation. Finally, Section 7 concludes the paper and summarises the key findings of the study.

2. A Statistical Approach to Evaluating Language Models

Evaluating the performance of language models is a key task in research on natural language processing. As language models get more complex and powerful, we need to find dependable ways to assess their performance. Traditional methods often use basic metrics like accuracy or perplexity. While these metrics give some insight into model performance, they don't always show the complete picture of how a model reacts to different data samples.

A statistical approach to evaluating language models tries to overcome these limitations by adding statistical analysis to the evaluation process. Instead of just looking at single performance values, statistical evaluation looks at the variability, reliability, and significance of model results. By using statistical techniques, researchers can find out if differences between models are significant or just random variations in the data.

Statistical evaluation methods are common in machine learning research. They offer a stricter comparison between models. These methods help confirm that conclusions about model performance are based on statistical evidence—not just on observed performance values. In this section, we will discuss the main statistical concepts used for evaluating language models in detail. These concepts include the evaluation process, performance metrics, statistical variability, confidence intervals, and hypothesis testing.

2.1 Evaluation of Language Models

Language models are usually judged by how well they can generate or predict meaningful text. The evaluation process starts with giving the model an input prompt, then looking at the text it produces. Depending on the task, this output is compared to a reference answer or the correct response.

For instance, in a question-answering task, the model gets a question as input and gives an answer. This answer is then checked against the expected answer to see if it is correct.

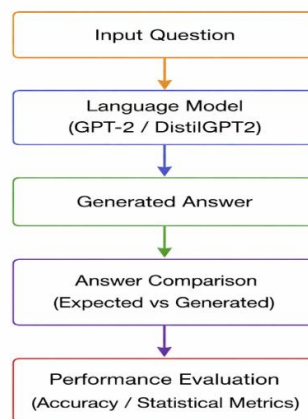


Figure 1: General evaluation pipeline for language models.

This evaluation process helps researchers measure a model's performance across different inputs. By repeating this on a dataset with many examples, it becomes possible to calculate statistics that summarise how well the model performs.

However, just counting the correct responses does not give a full picture of the model's behaviour. So, statistical evaluation techniques are used to look at the results in more detail.

2.2 Importance of Statistical Evaluation

Statistical evaluation is key in comparing machine learning models. When two models show similar accuracy, it can be hard to tell which is genuinely better.

For example, if one model has an accuracy of 0.80 and another 0.82, the second model may look slightly better. But this difference could be due to randomness in the dataset.

Statistical evaluation helps determine whether these differences are meaningful. By analysing the variability and distribution of the results, researchers can determine if the observed difference is statistically significant.

Table 2.1 Traditional vs Statistical Model Evaluation

Evaluation Type	Description
Traditional Evaluation	Uses single metrics such as accuracy or error rate
Statistical Evaluation	Uses statistical analysis to measure reliability and variability
Goal	Determine whether model differences are meaningful
Methods	Variance, standard deviation, confidence intervals, hypothesis testing

Using statistical evaluation methods offers several benefits:

- Researchers can measure how reliable a model's performance is.
- It helps identify if differences between models are statistically significant.
- It lowers the chance of making wrong conclusions based on experimental results.

2.3 Accuracy as a Basic Performance Metric

Accuracy is one of the most commonly used metrics for evaluating machine learning models. It measures the percentage of correct predictions made by a model.

Accuracy is calculated as:

$$\text{Accuracy} = (\text{Number of Correct Predictions}) / (\text{Total Number of Predictions})$$

In relation to language models, accuracy shows how often the model provides the correct answer when given a question.

While accuracy helps gauge overall performance, it has some limitations. It does not give information about how results vary across different data samples. Two models might have similar accuracy rates but behave differently with various inputs.

Due to these limitations, we need additional statistical measures to better understand model performance.

2.4 Variance and Standard Deviation

Variance is a statistical measure that explains how much a set of values differs from the average. In model evaluation, variance shows how much a model's performance fluctuates across different inputs.

A model with high variance may yield inconsistent results. For example, it may get some questions right while missing others.

Standard deviation is closely related to variance and indicates the average distance of values from the mean. It is often easier to understand than variance since it uses the same scale as the original data.

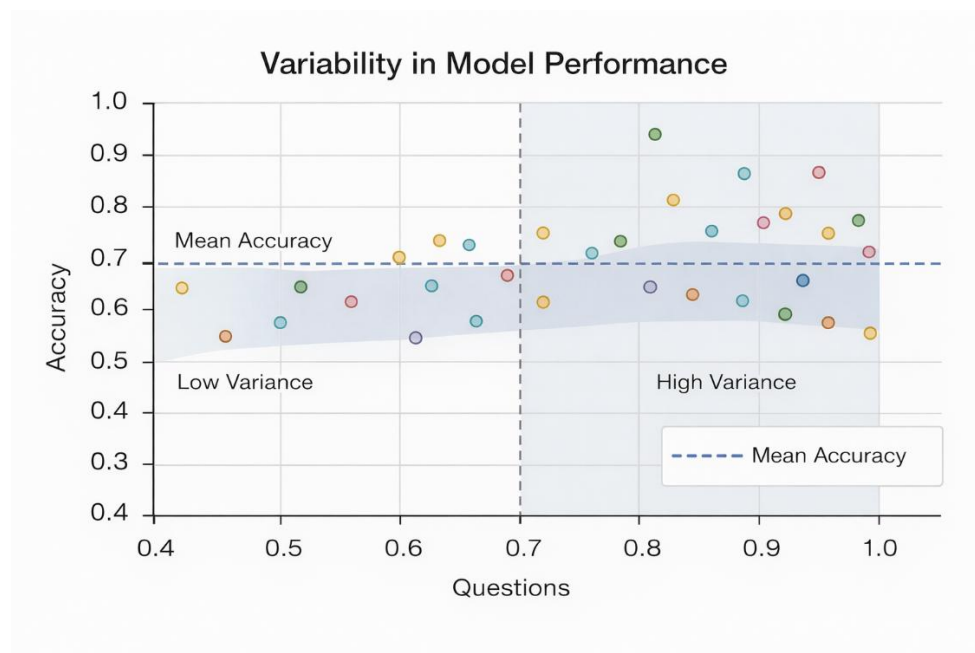


Figure 2: Example variation of model performance across different inputs.

Understanding variance and standard deviation is important for analysing the stability and consistency of model predictions. In machine learning evaluation, variance measures how much the model's performance varies across different inputs or data samples. If the variance is high, it means that the model's results fluctuate significantly. This suggests that the model

may perform well on some examples but poorly on others. Such behaviour indicates that the model's predictions are less stable and may rely heavily on the specific data used for evaluation. Analysing variance helps researchers see whether the model performs consistently across the dataset.

The standard deviation is closely related to the variance. It represents the average difference between the results and the mean performance. It offers a clearer interpretation of variability because it is measured in the same scale as the evaluation metric. A smaller standard deviation indicates that the model's performance is more consistent and that most results are near the average. In contrast, a larger standard deviation indicates greater variability in model predictions. By examining variance and standard deviation together, researchers can better assess the reliability of a model's performance and gain deeper insight into how stable the model is when handling different inputs.

2.5 Confidence Intervals

Confidence intervals provide a range of values for the likely true performance of a model. Rather than focusing on a single accuracy value, confidence intervals offer an estimate of the uncertainty tied to the evaluation results.

In machine learning evaluation, confidence intervals are often calculated using bootstrap sampling. This process involves repeatedly sampling from the dataset and calculating the evaluation metric for each sample. The distribution of these values can help estimate the range where the true performance of the model is likely to occur.

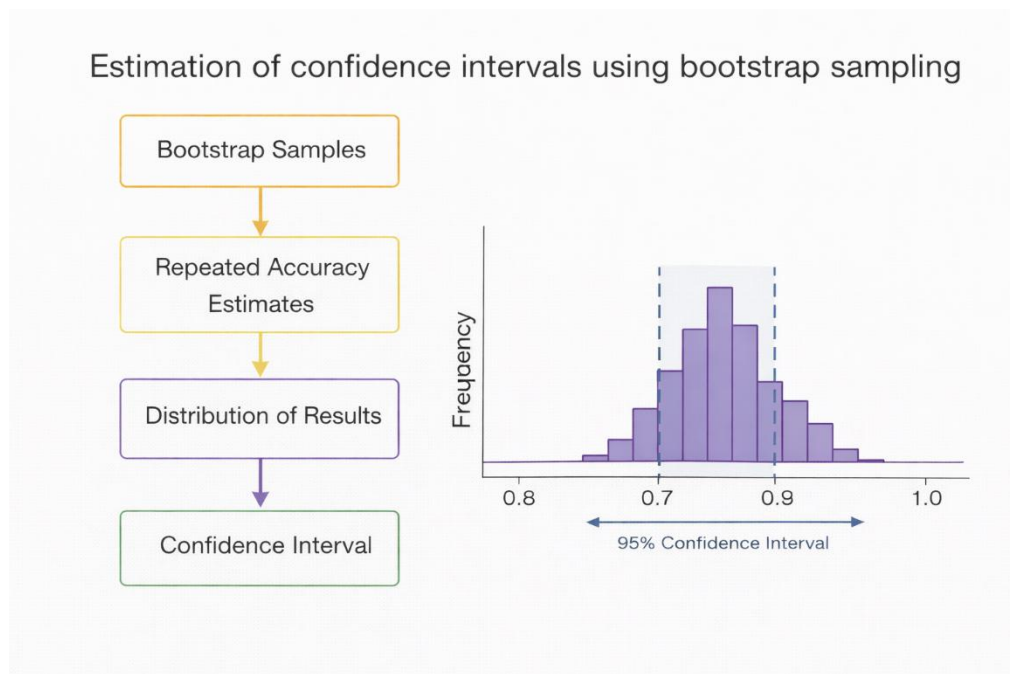


Figure 3: Estimation of confidence intervals using bootstrap sampling.

Figure X shows how to estimate confidence intervals using the bootstrap sampling method. This method generates several bootstrap samples from the original dataset by randomly

selecting data points with replacement. For each sample, the model's accuracy is computed, yielding a distribution of accuracy values. From this distribution, the lower and upper bounds of the confidence interval are found, typically representing a 95% confidence range. This interval shows where the true model performance is likely to fall and helps evaluate the reliability of the results.

2.6 Hypothesis Testing

Hypothesis testing is a statistical method for determining whether the difference between two models is statistically significant.

In hypothesis testing, we define two hypotheses:

Null Hypothesis (H_0)

The two models have the same performance.

Alternative Hypothesis (H_1)

The two models have different performances.

Next, we perform a statistical test to calculate a value called the p-value. The p-value shows the probability that the observed difference between the models happened by chance.

Table 2.2 Interpretation of hypothesis testing results (P-Values).

P-Value	Interpretation
$p < 0.05$	The difference between models is statistically significant
$p \geq 0.05$	The difference may be caused by randomness

Hypothesis testing helps researchers draw more reliable conclusions about how well a model performs.

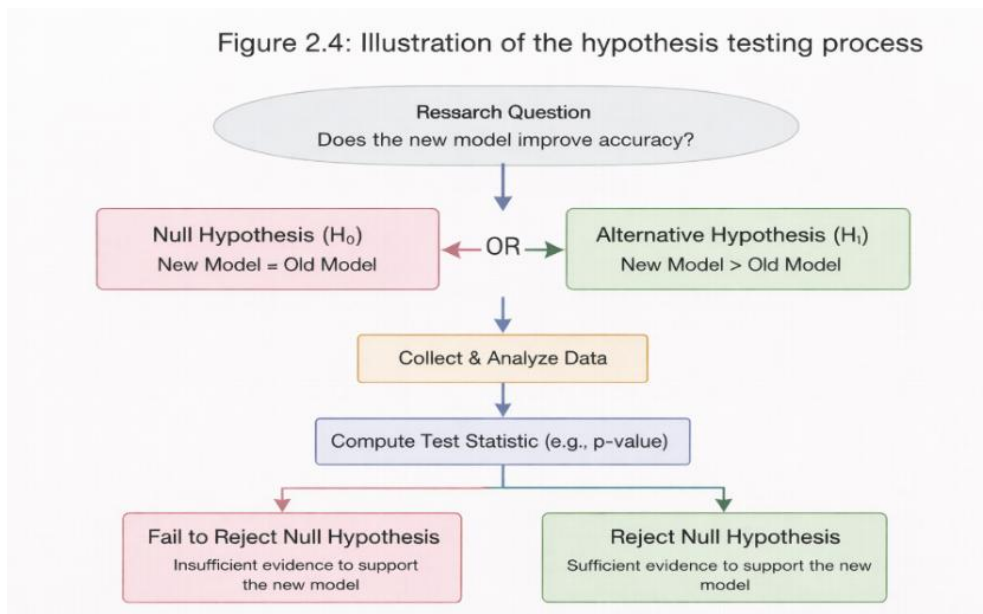


Figure 4: illustrates the hypothesis testing process

The figure above illustrates the process of hypothesis testing for evaluating statistical models. The process starts by defining a research question, such as whether a new model works better than an existing one.

From this question, two hypotheses are formulated: the null hypothesis (H_0) assumes no significant difference between the models, while the alternative hypothesis (H_1) suggests that one model is better than the other.

After collecting and analysing the evaluation data, a statistical test computes a test statistic, usually the p-value. Based on the p-value, a decision is made to either reject the null hypothesis, which indicates a significant improvement, or fail to reject the null hypothesis, meaning there is not enough evidence to claim improvement.

This process ensures that model comparisons rely on statistical evidence rather than just observed performance differences.

2.7 Summary

Using a statistical approach to evaluate language models offers a more reliable way to compare their performance. Rather than just looking at basic performance metrics like accuracy, statistical evaluation methods examine variability, reliability, and the significance of the results. Key statistical techniques for model evaluation include variance analysis, standard deviation, confidence interval estimation, and hypothesis testing.

These methods help researchers determine if the differences between models are meaningful and if the results are consistent across different data samples. By using statistical evaluation methods, researchers can make better choices when selecting or comparing machine learning models. The next sections of this paper apply these statistical concepts to compare the performance of GPT-2 and DistilGPT2 using an experimental evaluation framework.

3. Experimental Methodology

This chapter outlines the experimental setup used to test and compare the performance of two language models, GPT-2 and DistilGPT2. The goal of the experiment is to see if a smaller distilled model can perform as well as the original larger model. To do this, both models were tested on a dataset of factual questions, and their answers were assessed with statistical methods.

The experiment was designed to ensure a fair comparison between the models by using the same dataset, evaluation process, and statistical analysis methods. Various statistical metrics were employed in the evaluation, including accuracy, variance, standard deviation, confidence intervals, and hypothesis testing. These metrics provide a clearer understanding of the models' performance beyond just basic accuracy.

The experimental methodology includes several steps: model selection, dataset preparation, response generation, performance evaluation, and statistical analysis. Each step is detailed in the sections that follow.

3.1 Model Selection

Two language models were chosen for this experiment:

- GPT-2
- DistilGPT2

GPT-2 is a transformer-based language model developed by OpenAI. It generates coherent and contextually appropriate text. With many parameters and trained on a vast dataset of internet text, it performs well on many natural language processing tasks.

DistilGPT2 is a smaller version of GPT-2 created through a process called knowledge distillation. In this method, a smaller model (student model) learns to imitate the behaviour of a larger model (teacher model). The aim is to reduce the model size and computational requirements while maintaining similar performance levels.

Comparing GPT-2 and DistilGPT2 is valuable because it allows researchers to assess whether model compression techniques can reduce computational costs without significantly degrading performance.

Table 3.1: Comparison of Selected Models

Model	Type	Size	Description
GPT-2	Transformer Language Model	Large	Original model with high capacity
DistilGPT2	Distilled Transformer Model	Smaller	Compressed version of GPT-2

3.2 Dataset Preparation

To evaluate the models, we used a dataset of factual question-answer pairs. These questions cover general knowledge topics such as geography, science, history, and literature.

Here are some examples of the questions in the dataset:

- What is the capital of France?
- Who wrote the novel 1984?
- What is the largest ocean on Earth?
- Which planet is known as the Red Planet?

We created the dataset to make sure the questions were simple and factual. This way, we could easily check if the model responses were correct.

Each question was given to both models, and we collected the generated responses for analysis. Then we evaluated the responses to determine whether the model produced the correct answer.

3.3 Experimental Workflow

The experiment follows a clear workflow that keeps the evaluation process consistent and reproducible.

The main steps in the experimental workflow include:

1. Load the language models
2. Prepare the dataset of factual questions
3. Generate responses from each model
4. Evaluate the correctness of the responses
5. Calculate performance metrics
6. Apply statistical analysis.

This process ensures that both models are assessed under the same conditions.



Figure 3.1: Experimental Workflow of Language Model Evaluation

Figure 5: Experimental Workflow of Language Model Evaluation

3.4 Response Generation

We provided each question in the dataset as input to both language models. The models created text responses based on what they had learned.

The generation process included:

Tokenising the input question

Feeding the input into the language model

Generating a text response

Extracting the predicted answer

We used the same generation settings for both models to keep things consistent. This helps ensure that any performance differences come from the models themselves, not from different configurations.

3.5 Evaluation Metrics

We used several evaluation metrics to analyse how the models performed.

Accuracy:

Accuracy measures the proportion of questions the model answers correctly.

$$\text{Accuracy} = (\text{Number of Correct Predictions}) / (\text{Total Number of Questions})$$

Accuracy gives a basic measure of model performance.

Variance:

Variance shows how much the accuracy values change across different samples of the dataset. High variance means the model's performance is inconsistent.

Standard Deviation:

Standard deviation indicates the average deviation from the mean accuracy. A lower standard deviation means more stable model performance.

Confidence Intervals:

Confidence intervals provide an estimated range within which the true model accuracy is likely to fall. In this experiment, we calculated confidence intervals using bootstrap sampling.

Hypothesis Testing:

We used hypothesis testing to see if the difference between the two models is statistically significant.

The hypotheses for the experiment are:

Null Hypothesis (H_0):

There is no significant difference between GPT-2 and DistilGPT2.

Alternative Hypothesis (H_1):

There is a significant difference between the two models.

We calculated the p-value to decide whether to reject the null hypothesis.

3.6 Implementation Environment

We implemented the experiment using Python and ran it in Google Colab.

The following libraries were used:

- Transformers – for loading language models
- PyTorch – for deep learning operations
- NumPy – for numerical computations
- Matplotlib – for visualisation
- SciPy – for statistical analysis

Using these tools allows for efficient model evaluation and statistical analysis.

Table 3.2: Software and Tools Used

Tool	Purpose
Python	Programming language
Google Colab	Experiment environment
Transformers Library	Model loading
NumPy	Numerical calculations
Matplotlib	Graph visualization
SciPy	Statistical testing

3.7 Experimental Design

Considerations: Several factors were considered when designing the experiment to ensure reliable results.

First, the same dataset was used to evaluate both models, ensuring a fair comparison. Second, statistical methods were applied to look at the variability in the results. Finally, multiple evaluation metrics were used to better understand model performance.

Using statistical evaluation methods helps lower the risk of drawing incorrect conclusions based only on simple accuracy values.

4. Results and Analysis

This chapter presents the results from the experimental evaluation of the two language models, GPT-2 and DistilGPT2. The goal of the experiment was to see if a smaller distilled model could perform similarly to the original larger model. Both models were evaluated on the same factual question dataset, and their responses were evaluated using several statistical metrics.

The analysis compares accuracy, variance, standard deviation, confidence intervals, and hypothesis testing results. These statistical measures provide a clearer understanding of model performance and enable a reliable comparison between the two models.

4.1 Accuracy Comparison

Accuracy is one of the most used metrics for evaluating machine learning models. It measures the proportion of questions the model answers correctly. In this experiment, accuracy was calculated as the number of correct responses divided by the total number of questions in the dataset.

The results show that GPT-2 achieved higher accuracy than DistilGPT2. This indicates that the larger model answered more factual questions correctly.

Table 4.1: Accuracy Comparison of Language Models

Model	Accuracy
GPT-2	0.12
DistilGPT2	0.06

From the table, we can see that GPT-2 correctly answered about 12% of the questions, while DistilGPT2 achieved 6% accuracy. This suggests that the larger model performs better on the selected dataset.

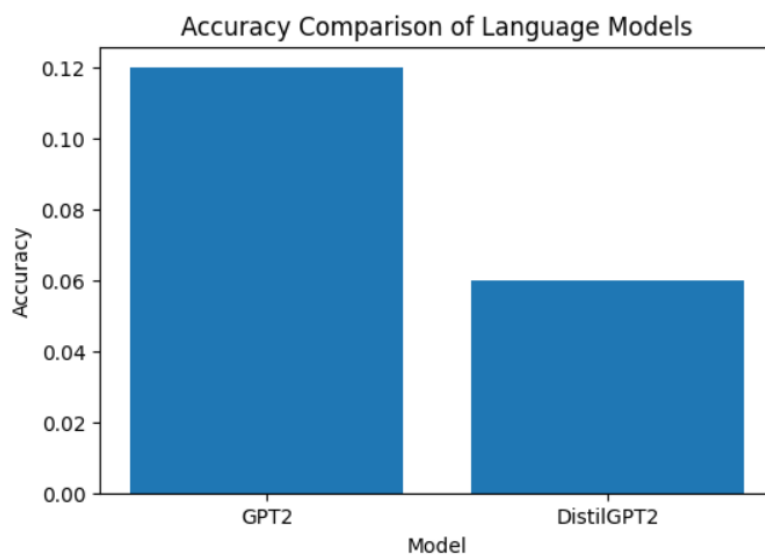


Figure 6: Accuracy Comparison Between GPT-2 and DistilGPT2

The figure above visually compares the accuracy of the two models. The bar chart clearly shows that GPT-2 performs better than DistilGPT2 in answering factual questions correctly.

4.2 Variance and Standard Deviation Analysis

In addition to accuracy, it is important to consider how variable the model's performance is. Variance and standard deviation are statistical measures that help us understand how consistent the model predictions are across different samples.

Variance measures how spread out the results are around the average accuracy. A higher variance indicates that the model's performance varies more across samples. Standard deviation is the square root of variance and gives a more understandable measure of variability.

Table 4.2: Variance and Standard Deviation of Model Performance

Model	Variance	Standard Deviation
GPT-2	0.1056	0.3249
DistilGPT2	0.0564	0.2375

The results show that GPT-2 has a slightly higher variance than DistilGPT2. This means that while GPT-2 achieves higher accuracy, its performance varies slightly more across different samples in the dataset.

4.3 Confidence Interval Estimation

Confidence intervals were calculated using bootstrap sampling. Bootstrap sampling involves repeatedly drawing samples from the dataset with replacement and calculating the model's accuracy for each sample. This process creates a distribution of accuracy values, which can help estimate the range in which the true model accuracy is likely to fall.

The calculated confidence intervals indicate how reliable the model's performance is.

Table 4.3: Confidence Interval Estimation

Model	Lower Bound	Upper Bound
GPT-2	0.04	0.20
DistilGPT2	0.00	0.14

The confidence interval for GPT-2 ranges from 0.04 to 0.20. The interval for DistilGPT2 spans from 0.00 to 0.14. Although the confidence intervals show some overlap, the hypothesis testing results in the next section indicate that the difference between the two models is significant.

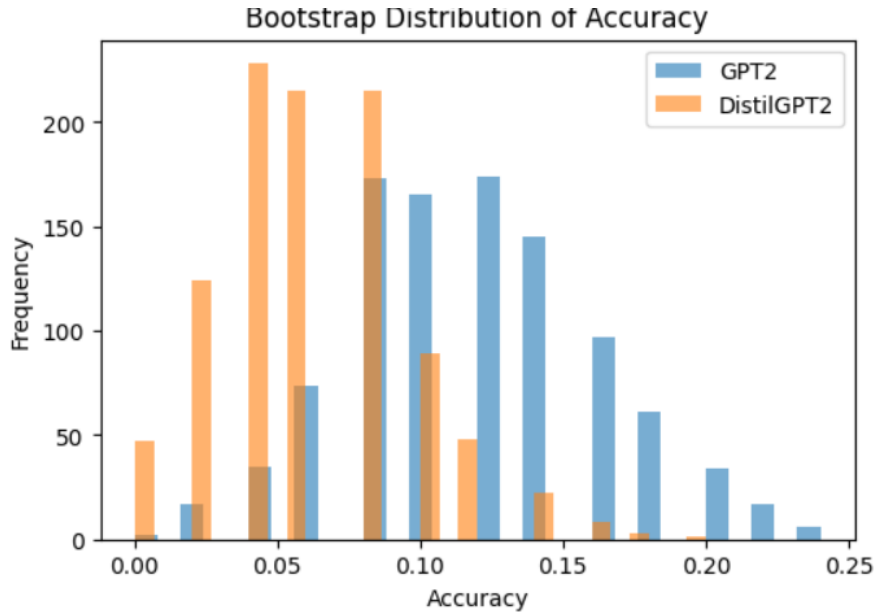


Figure 7: Bootstrap Distribution of Model Accuracy

The figure shows the distribution of accuracy values generated through bootstrap sampling. The spread of the distribution reflects the variability of the model's performance across different samples.

4.4 Hypothesis Testing

To further analyse the difference between the two models, we performed statistical hypothesis testing. This testing helps determine whether the observed difference in accuracy is statistically significant or due to chance.

The hypotheses used in this experiment are:

Null Hypothesis (H_0):

There is no significant difference between the performance of GPT-2 and DistilGPT2.

Alternative Hypothesis (H_1):

There is a significant difference between the performance of the two models.

We conducted a statistical test to compute the p-value, which indicates the probability that the observed difference occurred randomly.

Table 4.4: Hypothesis Testing Results

Model	P-value
GPT-2 vs DistilGPT2	0.0000

The p-value from the statistical test is very small, suggesting that the difference in accuracy between the two models is significant. This indicates that the performance difference observed in the experiment is unlikely to be due to random variation in the dataset.

4.5 Discussion of Results

The experiment's results show that GPT-2 achieves higher accuracy than DistilGPT2 on the factual-questions dataset. This outcome is expected because GPT-2 is a larger model with more parameters, giving it a better ability to learn complex language patterns.

DistilGPT2, in contrast, is a smaller version of GPT-2 designed to reduce computational costs and improve efficiency. While this compression reduces the model's size and increases speed, it may also cause a slight performance drop.

The results also emphasise the need for statistical evaluation. Rather than relying solely on accuracy values, the experiment included confidence intervals and hypothesis testing to provide a more reliable interpretation of the results.

5. Conclusion and Future Work

5.1 Conclusion

This term paper examined a statistical method for assessing language models by comparing two models: GPT-2 and DistilGPT2. The main goal was to see if a smaller distilled model could perform as well as the original larger model. To answer this question, both models were evaluated on a factual-question dataset using several statistical metrics.

The results showed that GPT-2 had higher accuracy than DistilGPT2 on the evaluation dataset. Even though DistilGPT2 is intended to be a smaller, more efficient version of GPT-2, the findings suggest that reducing the model size may lead to performance loss. The evaluation also included additional statistical measures, such as variance, standard deviation, confidence intervals, and hypothesis testing. These techniques provided a clearer picture of the models' performance beyond just accuracy comparisons.

The analysis of confidence intervals revealed the variability of the model results. Hypothesis testing was used to determine whether the difference in performance between the two models was statistically significant. The statistical analysis indicated that the difference in accuracy between GPT-2 and DistilGPT2 is significant. This suggests that the larger model performs better on the selected dataset.

Overall, the study emphasises the importance of using statistical methods when evaluating machine learning models. Rather than relying on a single metric such as accuracy, statistical evaluation provides a more reliable framework for comparing model performance. By examining variability, uncertainty, and statistical significance, researchers can gain a clearer understanding of the strengths and weaknesses of different models.

5.2 Future Work

This study provides valuable insights into the statistical evaluation of language models, but there are opportunities to expand this work in future research. First, the experiment utilised a

relatively small dataset of factual questions. Future studies might evaluate the models using larger, more diverse datasets to gain a broader understanding of their performance.

Second, this study focused on just two language models, GPT-2 and DistilGPT2. Future research could broaden the comparison by including other language models, such as GPT-Neo, BERT-based models, or different transformer architectures. Comparing multiple models would offer a wider perspective on how model size and structure affect performance.

Another potential extension of this work is to investigate additional evaluation metrics beyond those used in this study. Metrics such as precision, recall, F1-score, and semantic similarity measures could provide more insight into the quality of model responses.

Finally, future research could explore improvements to model distillation techniques. Since the goal of distillation is to create smaller, more efficient models while maintaining high performance, enhancing these methods could help reduce the performance gap between large models and their compressed counterparts.

In summary, statistical evaluation is crucial for understanding and comparing the performance of language models. By using rigorous statistical methods, researchers can ensure that model comparisons are meaningful, reliable, and evidence-based.

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Appendix

```
# Installing required libraries
# transformers → used to load pretrained language models
# torch → backend used by HuggingFace models
# numpy → numerical calculations
# matplotlib / seaborn → visualization of results
# scipy → statistical testing

!pip install transformers
!pip install torch
!pip install numpy
!pip install matplotlib
!pip install seaborn
!pip install scipy
```

```

from transformers import pipeline
import numpy as np
import random
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import ttest_ind
import pandas as pd

# Load language models
gpt2 = pipeline("text-generation", model="gpt2")
distil = pipeline("text-generation", model="distilgpt2")

print("Both models were loaded successfully.")

# Dataset of questions
questions = [
    "What is the capital of Germany?",
    "What is the capital of France?",
    "What is the capital of Italy?",
    "What is the capital of Spain?",
    "What is the capital of Canada?",
    "What is the capital of Japan?",
    "What is the capital of Australia?",
    "What is the capital of India?",
    "What is the capital of Brazil?",
    "What is the capital of China?",
    "Who wrote the novel 1984?",
    "Who wrote Hamlet?",
    "Who wrote Pride and Prejudice?",
    "Who wrote The Hobbit?",
    "Who painted the Mona Lisa?",
    "Who discovered penicillin?",
    "Who discovered gravity?",
    "Who invented the telephone?",
    "Who developed the theory of relativity?",
    "Who was the first president of the USA?",

```

"What is the largest ocean?",
 "What is the smallest continent?",
 "What is the tallest mountain?",
 "What is the longest river?",
 "What is the fastest land animal?",
 "What is the chemical symbol for gold?",
 "What is the boiling point of water?",
 "What gas do plants absorb?",
 "What planet is known as the Red Planet?",
 "What is the largest mammal?",
 "What is the currency of Japan?",
 "What is the currency of the UK?",
 "What is the square root of 64?",
 "What is the speed of light?",
 "What is the capital of Russia?",
 "What is the capital of Mexico?",
 "What is the capital of South Korea?",
 "What is the capital of Egypt?",
 "What is the capital of Argentina?",
 "What is the capital of Turkey?",
 "What is the largest desert?",
 "What is the tallest building?",
 "What is the largest planet?",
 "What is the smallest prime number?",
 "What year did World War II end?",
 "What year did World War I start?",
 "What is the freezing point of water?",
 "What is the largest island?",
 "What language is spoken in Brazil?",
 "What is the capital of Portugal?"

]

Correct answers

answers = [

"Berlin", "Paris", "Rome", "Madrid", "Ottawa", "Tokyo", "Canberra", "New Delhi", "Brasilia", "Beijing",
 "George Orwell", "Shakespeare", "Jane Austen", "Tolkien", "Da Vinci", "Fleming", "Newton", "Bell", "Einstein", "Washington",
 "Pacific", "Australia", "Everest", "Nile", "Cheetah", "Au", "100", "Carbon dioxide", "Mars", "Blue whale",

```
"Yen", "Pound", "8", "299", "Moscow", "Mexico City", "Seoul", "Cairo", "Buenos Aires", "Ankara",
"Sahara", "Burj Khalifa", "Jupiter", "2", "1945", "1914", "0", "Greenland", "Portuguese", "Lisbon"
]
```

```
# Model evaluation function
```

```
def evaluate(model, model_name):
    scores = []
    for question, answer in zip(questions, answers):
        prompt = f"Question: {question} Answer:"
        output = model(
            prompt,
            max_new_tokens=15,
            temperature=0.2,
            do_sample=True,
            top_k=40
        )[0]["generated_text"]
        generated = output[len(prompt):].strip()
        correct = 1 if answer.lower() in generated.lower() else 0

        print("\n-----")
        print("MODEL:", model_name)
        print("QUESTION:", question)
        print("MODEL OUTPUT:", generated)
        print("EXPECTED ANSWER:", answer)
        print("CORRECT:", correct)

        scores.append(correct)

    return scores
```

```
# Evaluate both models
```

```
gpt2_scores = evaluate(gpt2, "GPT2")
distil_scores = evaluate(distil, "DistilGPT2")
```

```
# Accuracy calculation
```

```
gpt2_accuracy = np.mean(gpt2_scores)
distil_accuracy = np.mean(distil_scores)
```

```

print("GPT2 Accuracy:", gpt2_accuracy)
print("DistilGPT2 Accuracy:", distil_accuracy)

# Accuracy comparison plot
models = ["GPT2", "DistilGPT2"]
accuracy = [gpt2_accuracy, distil_accuracy]

plt.figure(figsize=(6,4))
plt.bar(models, accuracy)

plt.title("Accuracy Comparison of Language Models")
plt.ylabel("Accuracy")
plt.xlabel("Model")
plt.show()

# Variance and standard deviation
gpt2_variance = np.var(gpt2_scores)
distil_variance = np.var(distil_scores)

gpt2_std = np.std(gpt2_scores)
distil_std = np.std(distil_scores)

print("GPT2 Variance:", gpt2_variance)
print("DistilGPT2 Variance:", distil_variance)

print("GPT2 Standard Deviation:", gpt2_std)
print("DistilGPT2 Standard Deviation:", distil_std)

# Plot accuracy with error bars
models = ["GPT2", "DistilGPT2"]
accuracy = [gpt2_accuracy, distil_accuracy]
errors = [gpt2_std, distil_std]
plt.figure(figsize=(6,4))
plt.bar(models, accuracy, yerr=errors, capsize=5)

plt.title("Model Accuracy with Standard Deviation")
plt.ylabel("Accuracy")

```

```

plt.xlabel("Model")
plt.show()

# Bootstrap sampling function
def bootstrap(scores, n=1000):
    means = []
    for _ in range(n):
        sample = random.choices(scores, k=len(scores))
        means.append(np.mean(sample))
    return means

# Bootstrap distributions
gpt2_boot = bootstrap(gpt2_scores)
distil_boot = bootstrap(distil_scores)

# Confidence intervals
print("GPT2 Confidence Interval:", np.percentile(gpt2_boot,[2.5,97.5]))
print("DistilGPT2 Confidence Interval:", np.percentile(distil_boot,[2.5,97.5]))

# Histogram of bootstrap distribution
plt.figure(figsize=(6,4))

plt.hist(gpt2_boot, bins=30, alpha=0.6, label="GPT2")
plt.hist(distil_boot, bins=30, alpha=0.6, label="DistilGPT2")

plt.legend()
plt.title("Bootstrap Distribution of Accuracy")
plt.xlabel("Accuracy")
plt.ylabel("Frequency")

plt.show()

# Kernel density estimation plot
plt.figure(figsize=(6,4))

sns.kdeplot(gpt2_boot, label="GPT2")
sns.kdeplot(distil_boot, label="DistilGPT2")

```

```

plt.legend()
plt.title("Confidence Distribution of Model Accuracy")
plt.xlabel("Accuracy")
plt.show()

# Hypothesis testing
t_stat, p_value = ttest_ind(gpt2_boot, distil_boot)
print("T-statistic:", t_stat)
print("P-value:", p_value)

# Compute confidence interval values
gpt2_ci_lower, gpt2_ci_upper = np.percentile(gpt2_boot,[2.5,97.5])
distil_ci_lower, distil_ci_upper = np.percentile(distil_boot,[2.5,97.5])

# Create results table
results_table = pd.DataFrame({
    "Model": ["GPT2", "DistilGPT2"],
    "Accuracy": [gpt2_accuracy, distil_accuracy],
    "Variance": [gpt2_variance, distil_variance],
    "Std Deviation": [gpt2_std, distil_std],
    "Confidence Interval Lower": [gpt2_ci_lower, distil_ci_lower],
    "Confidence Interval Upper": [gpt2_ci_upper, distil_ci_upper],
})
results_table

# Add p-value
results_table["P-value"] = [p_value, p_value]
results_table

# Format table output
results_table.style.format({
    "Accuracy": "{:.3f}",
    "Confidence Interval Lower": "{:.3f}",
    "Confidence Interval Upper": "{:.3f}",
    "P-value": "{:.4f}"
})

```

Affidavit /Declaration of Academic Integrity

I hereby confirm that my thesis entitled is the result of my own work. I did not receive any help or support from commercial consultants. All sources and/or materials applied are specified, referenced and listed in the thesis.

Furthermore, I confirm that this thesis has not yet been submitted as part of another examination process, neither in identical nor in similar form.

Date :06-03-2026**Signature :**Samyuktha Police